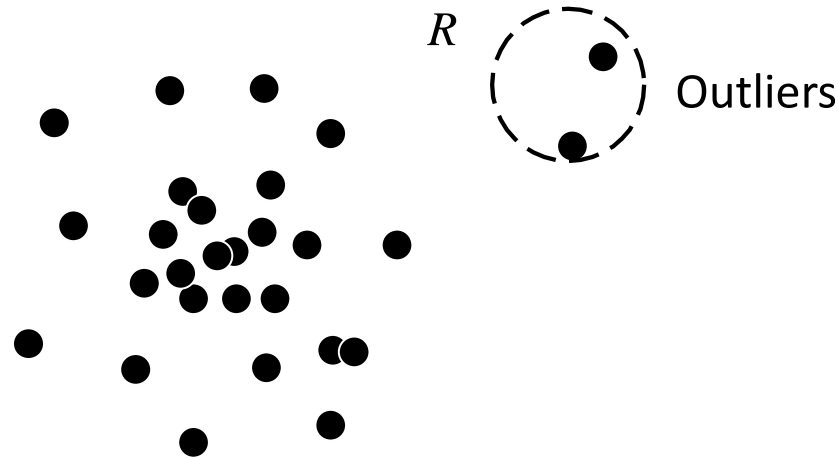


Module 9: Outlier Detection

- ❑ Basic concepts
 - ❑ What are outliers?
 - ❑ Types of outliers
 - ❑ Challenges in outlier detection
- ❑ Statistical approaches
- ❑ Proximity-based approaches
- ❑ Reconstruction-based approaches
- ❑ Clustering and classification-based approaches
- ❑ Mining contextual and collective outliers
- ❑ Outlier detection in high-dimensional data

What Are Outliers?

- ❑ Assume that a given statistical process is used to generate a set of data objects
- ❑ An outlier is a data object that deviates significantly from the rest of the objects, as if it were generated by a different mechanism



Outliers versus Noise

- ❑ Noise is a random error or variance in a measured variable, and thus is not interesting in data analysis, including outlier detection
 - ❑ Example: a bigger lunch one day or one more cup of coffee than usual
- ❑ Outliers are interesting because they are suspected of not being generated by the same mechanism as the rest of the data
 - ❑ It is important to justify why the detected outliers are generated by some other mechanisms
 - ❑ Example: make various assumptions on the rest of the data and show that the detected outliers violate those assumptions significantly

Outlier Detection and Novelty detection

- ❑ Example: monitoring a social media web site where new content is incoming, novelty detection may identify new topics and trends in a timely manner
 - ❑ Novel topics may initially appear as outliers
- ❑ To this extent, outlier detection and novelty detection share some similarity in modeling and detection methods.
- ❑ Difference: in novelty detection, once new topics are confirmed, they are usually incorporated into the model of normal behavior so that follow-up instances are not treated as outliers anymore